

Valentin Gaucher

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🎓 Google Scholar

in Valentin Gaucher

About Me

Robotist and aerospace engineer working on data-driven and nonlinear MPC for aerial robots in contact-rich, interactive environments. I leverage advances in computation and embedded electronics to build robotic systems that are safer, smarter, and more agile. **Anticipated Graduation: Spring 2028. French citizen, F-1, CPT-eligible.**

Education

PhD Arizona State University (USA) , Optimal Control & Robotics	Aug. 2024 to Spring. 2028
M.T California Institute of Technology (USA) , Master Thesis, Aerospace & Robotics	Apr. 2023 to Apr. 2024
M.S ISAE Supaero (France) , Aerospace Engineering, Dynamics & Control	Sep. 2021 to Sep. 2023
V.S Politecnico di Milano (Italy) , Visiting Student, Aerospace & Control	Feb. 2021 to Sep. 2021
B.S ENSTA (France) , Mechanical Engineering	Sep. 2018 to Sep. 2021

Additional Training

ETH Zürich , Autonomous Mobile Robots Online Course (Computer Vision, SLAM)	Jan. 2024, 6 months
NASA Jet Propulsion Laboratory , DARTS Lab Workshop (Flexible Multibody Dynamics)	Aug. 2023, 1 week

Relevant Experience

Robotics and Intelligent Systems Laboratory (RISE), ASU , Graduate Research Assistant <i>Developing control algorithms to solve trajectory optimization problems in drone-environment interactions: Data Driven and Non-Linear MPC, hybrid dynamical systems, optimal control for UAVs in contact-rich applications.</i>	Tempe, USA Aug. 2024 to Now Current
Centre National d'Etude Spatiales (CNES) , Mars Rover Operations Engineer <i>Real-time operation and commanding of ChemCam (MSL) and SuperCam (Mars 2020) science instruments aboard active Mars rovers. Responsible for uplink sequence validation, instrument health monitoring, and anomaly response in a fault-critical planetary robotics environment at the CNES Toulouse Space Center (French Space Agency).</i>	Toulouse, France April 2024 to August 2024 6 Months
NASA Jet Propulsion Laboratory / Caltech CAST , Robotics, Visiting Student Researcher <i>Experimental characterization of Mars Science Helicopter takeoff performance from a jet-pack. Takeoff dynamics & control, development of a real-time robust controller. Testing at Caltech GALCIT CAST facility.</i>	Pasadena, USA Apr. 2023 to Nov. 2023 8 months
AerospacePoliMi , Control, Visiting Student <i>Quadrotor robust attitude controller using H_∞ and Monte Carlo validation. Studied aero-elasticity effects on quadrotors for robust control purposes.</i>	Milan, Italy Feb. 2021 to Aug. 2021 6 months

Research Interests

Aerial robotics for extreme environments: Deployment, sensing, and control of aerial robots in extreme environments. Investigating tactile based planning [6] and sensing methods for planetary exploration [4], sample recovery and search and rescue.

Interaction-Rich Optimal Control and Representation: Planning and control that let dynamical systems survive, leverage or recover from contact. This include collision recovery [1], contact-aware optimal planning [2][6], landing and takeoff [2][3], and contact modeling [1].

Real-time control and experimental validation: Deployment of real-time optimal control methods on flight hardware and high-fidelity simulation (MuJoCo, ROS2, PX4), with a focus on closing the simulation-to-real gap [3]–[7].

Research Software & Tools

MuJoCo-PX4 Simulation and Deployment Pipeline: Developed a MuJoCo-based simulation environment running the full ROS2-PX4 stack as an alternative to Gazebo, supporting configurable obstacle courses, vision sensors and IMU for rapid prototyping of optimal control algorithms. Currently extending it, with a supervised intern, as a deployment agent that automatically ports simulation-side controllers to the ROS2-PX4 stack for flight on real hardware. Used as the teaching platform for the graduate course *System Control and Optimization* at ASU.

Darvis — Natural Language Assistant for Drone Experiments: Originated and lead contributor of an LLM-based assistant for live ROS2-PX4 flight sessions, allowing operators to modify setpoints and trigger deployments by voice or text, extract and query flight logs, and diagnose failed runs in natural language.

Papers

[1] **Published - IEEE Control Systems Letters (L-CSS), 2026:** *Dual Quaternion Based Contact Modeling for Fast and Smooth Collision Recovery of Quadrotors*. Valentin Gaucher & Wenlong Zhang. IEEE link: <https://ieeexplore.ieee.org/document/11569027>

[2] **Accepted - IEEE Conference on Decision and Control (CDC), 2026:** *Switch-Time-Aware Hybrid Model Predictive Control Using the Hybrid Minimum Principle*. Valentin Gaucher & Wenlong Zhang.

[3] **Under Review - IEEE International Conference on Robotics and Automation (ICRA):** *Real Time Neural Model Predictive Control for Takeoff and Landing with Learned Ground Effect Scaling*. Valentin Gaucher, Yizhuang Garrard & Wenlong Zhang.

[4] **Under Review - IEEE Transactions on Field Robotics (T-FR):** *Dynamically Scaled Earth-Ambient Testing of Airload-Aware Takeoff for Mars Mid-Air Helicopter Deployment*. Marcel Veismann , Valentin Gaucher , Fernando Ruiz Vincueria , Peter Gavrilov , Luis Phillipe Tosi , Severin Schumacher, Filip Slezak, Katherine Tighe , Jacob Izraelevitz, Morteza Gharib , and Jeff Delaune.

[5] **Under Review - International Conference on Learning Representations (ICLR):** *Fast Strategy Solving for the Informed Player in Two-Player Zero-Sum Linear-Quadratic Differential Games with One-sided Information*. Mukesh Ghimire, Valentin Gaucher, Wenlong Zhang & Yi Ren.

[6] **In Preparation - IEEE Transactions on Robotics (T-RO) :** *Online Collision-Inclusive MPC for Quadrotors via Convex Relaxation of Hybrid Impact Law*. Valentin Gaucher & Wenlong Zhang.

[7] **In Preparation - IEEE Robotics and Automation Letters (RA-L) :** *Attitude-Aware Obstacle Avoidance for Agile Quadrotor Flight using Dual-Quaternion Model Predictive Control*. Valentin Gaucher & Wenlong Zhang.

Skills

Languages: Python (10 years experience), C/C++ (4 years experience), Matlab/Simulink (8 years experience)

Hardware: Jetson Thor, Jetson Orin Nano, Raspberry Pi 4/5, Arduino, PX4 Flight Controller

Software Libraries: Pytorch, ROS2, MuJoCo, Gazebo, Solidworks, CATIA, Abaqus

Optimization Libraries: acados, CasADi, OSQP, CVXPY, IPOPT, CUDA

Awards

- "Space Fellowship": Awards given by CNES and French Air/Space Force for promising talent in Space Sector: <https://www.fondation-ailesdefrance.fr/bourses-espace/precedentes-editions/bourses-espace-2023/>
- "ASU Fulton Fellowship": Fellowship Award, merit- and need-based awards that are funded by private donors, companies, community members, professional organizations, fees and the Fulton Schools of Arizona State University: <https://students.engineering.asu.edu/graduate/fellowships-funding/>.
- "ASU University Graduate Fellowship (UGF)": Fellowship Award, merit based award based on Academic achievements and records.